

Tom Sloan

Robotic Vision Engineer

Willing to Relocate

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[GitHub/Tom-Sloan](https://github.com/Tom-Sloan) | [LinkedIn/tom-sloan](https://www.linkedin.com/in/tom-sloan) | www.tom-sloan.com

Key Technologies

3D Perception & SLAM:	ORB-SLAM3, SLAM3R (Neural SLAM), Visual-Inertial Odometry, LiDAR, Depth Sensing
Computer Vision & AI:	PyTorch, TensorFlow, 3D Reconstruction, Skeletal Tracking, Thermal Imaging
XR & Digital Twins:	Unity, Augmented Reality (AR) Headsets, Real-time 3D Visualization, VR Streaming
UAV & Robotics:	DJI SDK, Autonomous Navigation, Indoor Mapping, Sensor Fusion (RGB, IMU, LiDAR)
Cloud & DevOps:	AWS, Docker, RabbitMQ, Jenkins, CloudFormation, EC2, Firebase
Programming:	Python, C/C++, JavaScript, React.js, Node.js, MATLAB, PowerShell

Education

Master of Applied Science in Electrical and Computer Engineering

Thesis: Indoor 3D Modeling Using Consumer Drones and Neural SLAM for Virtual Reality and a Cloud Architecture

Carleton University, Ottawa | **December 2025**

B.Eng. in Computer Systems Engineering (with Distinction)

Carleton University, Ottawa | 2022

Publications

Featured Publication - Presented at ICCE 2026 (IEEE conference held in conjunction with CES)

T. Sloan, B. Wallace, and R. Goubran, "**Drone-Based Indoor Mapping for Augmented Indoor 3D Modeling Using Neural Simultaneous Localization and Mapping (SLAM)**," in 2026 IEEE International Conference for Consumer Electronics (ICCE), February 2026, **paper accepted**.

Other Publications

- T. Sloan, B. Wallace and R. Goubran, "**System for Drone-Based Indoor Mapping for Augmented Reality**," 2025 *IEEE International Instrumentation and Measurement Technology Conference (I2MTC)*, Chemnitz, Germany, 2025, pp. 1-6
- T. Sloan, B. Wallace and R. Goubran, "**Impact of Image Resolution on Controlling Drones Using Remote VR Headset Visualization and a Cloud Architecture**," 2025 IEEE Sensors Applications Symposium (SAS), Newcastle, United Kingdom, 2025, pp. 1-6

Thesis Research (Code open sourced at <https://github.com/Tom-Sloan/WorldSystem>)

- Developed a modular, cloud-assisted SLAM framework enabling fully autonomous indoor mapping and navigation with a low-cost consumer drone (DJI Mini 3) generating digital twins.
- Developed and compared two distinct SLAM approaches: a traditional visual-inertial system using ORB-SLAM3 with RGB camera and IMU fusion, and a neural SLAM system using SLAM3R operating with RGB camera data alone.
- Designed a cloud-based architecture using Docker, RabbitMQ, and React.js to offload high-computation tasks (SLAM, 3D reconstruction) to remote servers, achieving near real-time 3D reconstruction at 21.06 FPS on NVIDIA RTX 4070 Ti.
- Trained the system in a custom Unity-based simulator and validated performance through real-world deployments, achieving trajectory accuracy with 6.61 cm RMSE on the Replica benchmark.
- Enabled high-fidelity map streaming to AR headsets for on-the-fly visualization and monitoring, demonstrating practical applications for indoor spatial capture.

Work Experience

Machine Learning Researcher, Carleton University *2018 to 2021 & 2023 to Present*

- Engineered contact-based sensor systems using pressure-sensitive mats, embedded sensors, and custom PCBs for real-time physiological monitoring (heart rate, respiration rate, body fluid flow) in older adults.
- Deployed multi-modal camera infrastructure integrating LiDAR, thermal imaging, RGB webcams, and privacy-preserving smart cameras for behavioral monitoring in hospital settings, managing time synchronization across heterogeneous sensors in an intelligent monitoring system.
- Collaborated cross-functionally with faculty, graduate students, and industry partners to synthesize research findings, publish peer-reviewed papers, and guide continuous improvement of health-monitoring and UAV-based mapping systems.

DevOps Engineer, Magnet Forensics *2022 to 2023*

- Worked in a small team environment using DevOps tools including Jenkins, Linux, Python, and PowerShell to manage thousands of software builds daily on dozens of on-premise servers.
- Led migration from on-premise to AWS cloud using CloudFormation and EC2, applying knowledge from AWS Certifications.

Spectrum Engineering Co-Op, Telesat Canada *2020 to 2021*

- Developed an alternative user interface and API in Python to interact with MATLAB, enabling direct use of satellite XML data without requiring expensive MATLAB add-ons.
- Wrote extensive tests and code to analyze satellite spectrum use, enabling further analysis of international regulations such as ITU regulation 1503.

Independent Projects

Speakeasy: Google Gemini, ByteDance HLLM, Google Speech-to-Text, React Native, Node.js

Smart Home: C, C++, PyTorch, CAD, PCB Design, AWS, Firebase

Smart Band Project: C, C++, Python, CAD, PCB Design, AWS, MATLAB

Portfolio Website: React, Redux, JavaScript, HTML, Sass

More projects at www.tom-sloan.com/projects

Certifications

AWS Certified Machine Learning Engineer (Associate)	November 2025
AWS Certified Security (Specialty)	October 2022
AWS Certified SysOps Administrator (Associate)	May 2022
AWS Certified Solutions Architect (Associate)	January 2022
AWS Certified Developer (Associate)	February 2022
AWS Certified Cloud Practitioner	July 2020